Reinforcement Learning-Based Energy-Aware Fault-Tolerant Routing in Underwater Acoustic Networks

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Abstract

Underwater Acoustic Sensor Networks (UASNs) are used for ocean monitoring, marine exploration, and environmental surveillance [1, 2]. However, because of their limited battery energy, the unreliability of the acoustic links, and due to node or link failures, packet loss takes place, energy consumption goes up, and the lifetime of the network is shortened. To solve these problems, this study presents an Energy-Aware Fault-Tolerant Routing (EAFTR) method based on Q-learning which takes into account the residual energy of neighbouring nodes, the quality of the link, node availability, and progress towards the destination while avoiding known failures. The approach also examines alternative forwarding paths in order to ensure reliable communication when a node or a link fails. The approach will be evaluated by means of the packet delivery ratio, energy consumption, network lifetime, end-to-end delay, throughput, and routing cost in order to find out whether combining energy-aware decisions, reinforcement learning, and fault tolerance leads to an improvement in underwater network performance.

I Introduction

A Background

Underwater Acoustic Sensor Networks (UWANs) are applied to ocean monitoring, pollution detection and marine exploration [1]. Data transmission in these networks is carried out by means of acoustic signals, but communication is difficult because of the limited bandwidth, the high propagation delay, the varying channel conditions and the limited battery energy [3]. As it is hard to replace or recharge the batteries used in underwater, energy management becomes essential. Efficient routing strategies have to choose appropriate forwarding nodes in order to conserve energy, ensure reliable communication, cope with node or link failures and increase the lifetime of the network [2].

B Challenges

Routing in UWANs is challenging because underwater nodes have limited battery energy, and repeatedly using the same nodes can shorten the network lifetime. Acoustic links may also be unreliable, affecting successful data transmission. Furthermore, node or link failures can interrupt communication and make existing routes unavailable. Selecting a next-hop node based only on its remaining energy or progress toward the destination may not always provide a suitable route. Therefore, an effective routing method should balance energy consumption, link quality, and destination progress while avoiding known failed nodes and links and selecting alternative routes whenever possible[2, 3].

C Existing Approaches

Different routing methods have been developed for UWANs to save energy and improve communication, but each has certain limitations.

(i) **Depth-Based and Geographic Routing:**
These methods select the next hop based on node depth, position, or progress toward the destination[4, 5]. They use local information to make forwarding decisions, which keeps the process simple. However, they may reach dead ends or select nodes with low energy or poor link quality, which can affect reliable packet delivery.

(ii) **Energy-Aware Routing:**

These methods consider the remaining battery energy of nodes to reduce energy consumption [1, 6] .This helps with energy consumption and extends the lifetime of the network however, if one mainly focuses on energy can cause slow advancement towards the destination or result in unreliable communication links. Also, collecting updated energy information from neighboring nodes can increase communication overhead and affect network performance.

(iii) **Fault-Tolerant Routing:**

These methods identify alternative routes incase of nodes or links fail. Yet, route recovery can increase delay and overhead, and a suitable alternative path may not always be available.

These limitations motivate EAFTR, which combines reinforcement learning with energy awareness, link quality, destination progress, and failure handling to select suitable next hop

D Problem Statement

This work focuses on finding energy-efficient and reliable routes in underwater acoustic networks. The system consists of nodes which have limited battery power and where the communication links can fail. Routing method considers node energy, link quality, progress toward the destination and failure of a node or a link when choosing the next hop. It also skips those nodes which are unable to meet the destination. It looks for alternative routes when failures take place and adheres to the energy constraints. The aim is to improve reliability and energy consumption while at the same time reducing delay and routing overhead.

E Objectives

The main objectives of this work are:

- To develop a Q-learning-based routing algorithm that considers node energy, link quality and advancement in reaching the destination.
- This involves dealing with failures of nodes and links and, where possible, finding alternative routes.
- To evaluate the approach using packet delivery ratio, energy consumption, network lifetime, it involves delay, throughput, packet loss, and routing overhead.

The rest of this paper is organised as follows: Section II looks at existing routing methods and related work, while Section III outlines the proposed EAFTR methodology and algorithm.

II Literature Review

Underwater Acoustic Sensor Networks (UWANs) face challenges such as limited energy, unreliable links and high communication delays. Previous research has looked into different methods of routing have been proposed to deal with these problems [1, 2]. This section looks at the main approaches, their working principles, and their limitations

A Depth-Based and Geographic Routing

Depth-based and geographic routing methods use node depth, position, or progress toward destination as a factor to select the next hop. They are quite simple since the forwarding decision are be based on local information, but these methods might end up in dead ends or choose the nodes that have inadequate energy or a low link quality.

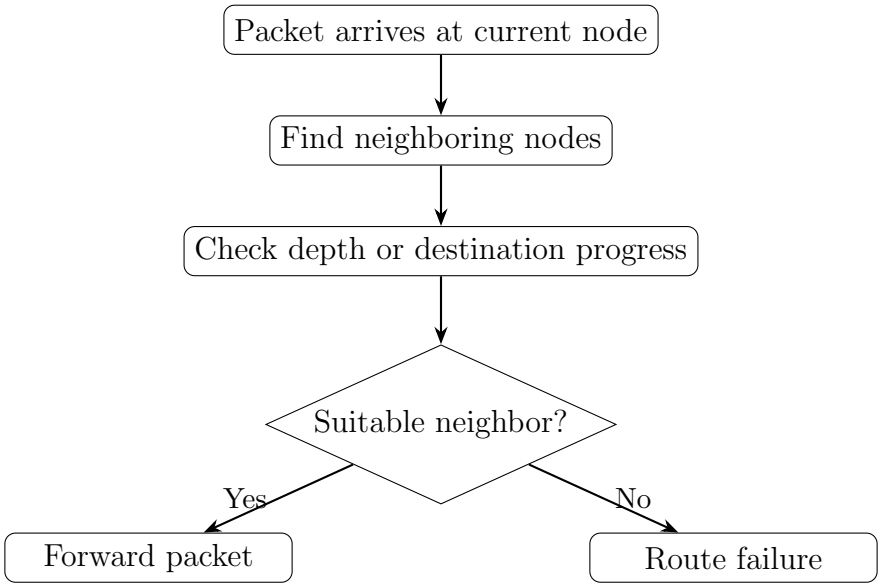


Figure 1: General forwarding process in depth-based or geographic routing.

Limitation: These methods may not consider residual energy or link reliability and greedy forwarding can fail when no suitable neighboring node is available.

B Energy-Aware Routing

When choosing a forwarding node, energy-aware routing takes the remaining battery energy or the estimated transmission cost into account, with the aim of reducing unnecessary energy consumption and avoiding the situation where some nodes exhaust their batteries too soon [1]. This in turn enables the energy demand to be spread out throughout the network and results in an increased total lifetime.

Algorithm 1 Simplified Energy-Aware Routing

Require: Current node, neighboring nodes, energy threshold

Ensure: Selected next-hop node or route failure

```
Find available neighboring nodes
for all neighbor v do
    if E_v is below the required energy threshold then
        Exclude v
    end if
end for
if no eligible neighbor remains then
    Report route failure
else
    Select an eligible neighbor based on residual energy
    Forward the packet to the selected neighbor
end if
```

Limitation: A node with more energy may have a poor link or make little progress toward the destination. Therefore, residual energy alone may not lead to the best routing decision.

C Fault-Tolerant Routing

The objective of fault-tolerant routing is to ensure that communication can continue even if a node or a link fails. On detection of a failure, the routing mechanism looks for another possible path to the destination. This is especially important in underwater networks since the communication links are unreliable and the nodes have limited energy. It therefore helps to decrease packet loss and enhances the overall reliability of data transmission.

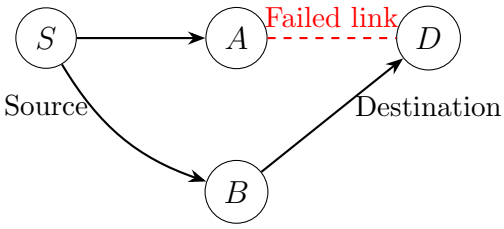


Figure 2: Illustration of an alternative route after a link failure.

Limitation: Failure detection and route recovery can increase the delay and control overhead.An alternative route may be unavailable if the network is disconnected or the other nodes cannot pass on the packet.

D Research Gap and Motivation

The approaches discussed above address different routing challenges, but relying on a single factor may lead to poor decisions. For example, advancement towards the destination does not guarantee sufficient energy, while the fact that there is a lot of residual energy does not ensure that the connection will be reliable. Similarly, fault recovery may fail when the alternative path is not feasible.Previous research has explored Q-learning for energy-aware and topology-aware underwater routing [7]. Recent approaches have also considered service requirements and routing-void handling [8].

To address these concerns, this work investigates EAFTR, a Q-learning-based routing approach that considers residual energy, link quality, destination progress, and forwarding feasibility is also taken into account along with the node or link fail to select the next hop. The method will be evaluated under different network conditions using packet delivery ratio, energy consumption, network lifetime, end-to-end delay, throughput, packet loss, and routing overhead

III Proposed Methodology

This work proposes Energy-Aware Fault-Tolerant Routing (EAFTR), which uses Q-learning to select suitable next-hop. The nodes in underwater acoustic networks considers energy, link quality, progress toward the destination, and takes into account of any known failures of nodes or links in order to improve routing decisions.

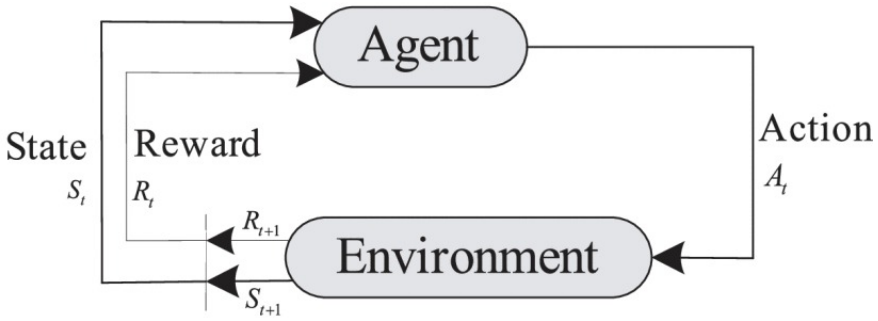


Figure 3: Reinforcement learning-based routing.

A Q-Learning-Based Routing

Q-learning allows a routing agent to improve its decisions through rewards. The agent learns the value of opting a particular next hop and updates its Q-values based on the reward that is received right away and that reward is anticipated in the future. The Q-learning update is given by:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right]$$

Here, s is the current state, a is the selected action, r is the immediate reward, and s' is the next state. The learning rate is represented by α, while γ controls the importance of future rewards.

B Reward Function

The reward function helps the agent choose suitable forwarding nodes by considering energy, link reliability, and progress toward the destination, while penalizing energy consumption and unnecessary leaps. It is given like this:

$$\text{Reward} = \text{Battery Benefit} + \text{Link Reliability} + \text{Destination Progress} - \text{Energy Cost} - \text{Hop Penalty}.$$

Battery benefit favors nodes with sufficient remaining energy, while link reliability favors better communication links: Destination progress promotes the forwarding of packets nearer to the destination. Energy cost and hop penalty encourage nodes not to use excessive energy or take additional hops. Reaching the destination can provide an additional positive reward. Together, these factors assist the agent in achieving a balance progress in the areas of energy use, reliability, and routing.

C Forwarding Feasibility and Failure Handling

Before choosing the next hop, EAFTR checks neighboring nodes and excludes visited nodes, known failed nodes or links and nodes that do not meet the assumed energy requirements also verifies whether a feasible path exists. If a node is inappropriate then it is taken out of the available actions. When a failure occurs in the network model, another feasible route may be chosen if there is one.

D Proposed Algorithm

Algorithm 2 EAFTR Using Q-Learning

Require: Network topology, node energies, failed nodes and links

Ensure: Learned Q-table

```
Initialize Q-values and learning parameters
for each training episode do
    Set current node to the source
    Initialize visited nodes
    while destination is not reached do
        Find feasible neighboring nodes
        if no feasible neighbor exists then
            End the episode
        else
            Select a next hop using an ϵ-greedy policy
            Calculate the reward and observe next state
            Update the Q-value
            Move to the selected next hop
            Update visited nodes
        end if
    end while
end for
Use learned Q-values for routing decisions
```

E Justification

In underwater acoustic networks routing is difficult because of the limited battery energy, the unreliable links and failures of nodes or links. It may not be possible to guarantee reliable and energy-efficient communication if the next hop is chosen solely on the basis of distance or residual energy. Hence, EAFTR makes use of Q-learning in order to learn appropriate forwarding decisions according to the rewards obtained during training.
The factors considered in the proposed approach are justified as follows:

- **Residual Energy:** Selecting nodes with sufficient remaining energy helps avoid overusing weak nodes and may improve network lifetime.
- **Link Quality:** Considering link quality helps avoid unreliable links and it could decrease packet loss and the need for retransmissions.
- **Destination Progress:** Encouraging progress toward the destination helps in avoiding unnecessary forwarding and reduce the number of hops.
- **Failure Handling:** Excluding known failed nodes and links and checking route feasibility helps the algorithm select a different option if one exists.
- **Q-Learning:** Q-learning allows the routing agent to update its decisions based on immediate and future rewards, instead of using a fixed rule for selecting the next hop.

The EAFTR approach is designed to achieve a balance between energy consumption, the reliability of communication, routing progress, and its effectiveness is assessed by means of simulations which use the packet delivery ratio, network lifetime, end-to-end delay, throughput, and routing overhead.

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